

Program: Diploma in Food Processing Technology	
Course Code: 5423A	Course Title: Food chemistry II
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4(L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

1. To understand the physiochemical changes in components of food
2. To acquire an understanding about food allergens and toxins formed
3. To impart knowledge on the effect of toxic substances on consumers/handlers

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of food chemistry	Food chemistry I

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the processes and applications of starch hydrolysis, dextrin and dextran production, fat modification techniques such as hydrogenation and inter-esterification, and the use of fat replacements.	17	Understanding
CO2	Describe the types of food allergies, the chemistry of food allergens, mechanisms of allergic reactions, and allergenic components in foods.	13	Understanding
CO3	Explain the methods for quantitative and qualitative analysis of toxicants in foods, biological determination of toxicants, and assessment of food safety, including risk assessment, acute toxicity, and various toxicological effects.	17	Understanding
CO4	Describe toxic substances in food and their toxic effects in people	13	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the processes and applications of starch hydrolysis, dextrin and dextran production, fat modification techniques such as hydrogenation and inter-esterification, and the use of fat replacements.		
M1.01	Explain dextrins and dextrans, Starch hydrolysis,	6	Understanding
M1.02	Discuss Modification of fats	6	Understanding
M1.03	Discuss fat replacements	5	Understanding
Contents: Dextrins and dextrans, Starch hydrolysis, Modification of fats: hydrogenation-cis and trans isomers, inter-esterification, Fat replacements.			
CO2	Describe the types of food allergies, the chemistry of food allergens, mechanisms of allergic reactions, and allergenic components in foods.		
M2.01	Discuss the chemistry and types of food allergens.	4	Understanding
M2.02	Explain the Mechanisms of allergic-type reactions	4	Understanding
M2.03	Illustrate food allergy and hypersensitivity,	4	Understanding
	Series Test –I	1	
Contents: Types, Chemistry of food allergens, Food allergy and hypersensitivity, mechanisms of allergic-type reactions and allergenic components in foods			

CO3	Explain the methods for quantitative and qualitative analysis of toxicants in foods, biological determination of toxicants, and assessment of food safety, including risk assessment, acute toxicity, and various toxicological effects.		
M3.01	Discuss quantitative and qualitative analysis of toxicants in foods	8	Understanding
M3.02	Explain biological determination of toxicants	6	Understanding
M3.03	Summarize assessment of food safety	3	Understanding
Contents: Quantitative and qualitative analysis of toxicants in foods; Biological determination of toxicants, Assessment of food safety – Risk assessment and risk benefit indices of human exposure, acute toxicity, mutagenicity and carcinogenicity, reproductive and developmental toxicity, neurotoxicity and behavioral effect, immunotoxicity			
CO4	Describe toxic substances in food and their toxic effects in people		
M4.01	Discuss Food additives as toxicants in food	4	Understanding
M4.02	Explain toxicants formed during food processing	4	Understanding
M4.03	Outline the risk of genetically modified food, food supplements, persistent organic pollutants	4	Understanding
	Series Test–II	1	
contents: Food additives as toxicants: artificial colors, preservatives, sweeteners; toxicants formed during food processing such as nitrosamines, maillard reaction products acrylamide, benzene, and irradiation; risk of genetically modified food, food supplements, persistent organic pollutants,			

Text / Reference

T/R	Book Title/Author
R1	Concon JM.1988. Food Toxicology - Principles & Concepts. Marcel Dekker.
R2	Steven T. 1989. Food Toxicology: A Perspective on Relative Risks
R3	Helferich, William and Carl K.Winter “Food Toxicology” CRC Press, 2001
R4	Shibamoto, Taka yuki and Leonard F.Bjeldanzes “Introduction to Food Toxicology” 2nd Edition.Academic Press, 2009.

R5	Shibamoto T. and Bjeldanes L., Introduction to Food Toxicology, Academic Press, Inc. San Diego, CA. ISBN 0-12640025-3 2. Deshpande, S.S. (2002).
R6	Handbook of Food Toxicology, Marcel Dekker Inc. NY ISBN 0- 8247-0760-5
R7	Stine, K. and Brown, T. 1996. Principles of Toxicology. CRC Press, Inc. Boca Raton, FL.
R8	Püssa, Tõnu. 2007. Principles of Food Toxicology. CRC Press, LLC. Boca Raton, FL. 5.

Online Resources

Sl.No	Website Link
1	https://foodscience.ucdavis.edu/academic-programs/undergraduate/courses/fst-128-food-toxicology
2	https://www.foodallergy.org/living-food-allergies/food-allergy-essentials/common-allergens
3	https://www.fda.gov/food/chemical-contaminants-pesticides/natural-toxins-food
4	https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food
5	https://www.futurelearn.com/info/courses/nutrition-for-health/0/steps/213600